The Foundations Canon

Ideas that changed how we understand computation, learning and intelligence. Each line: the work, then the idea it opened.

★ = start here (the first ten). free = freely available from the author or publisher.

1. Information and computation

★ Shannon (1948), *A Mathematical Theory of Communication*: information is measurable; prediction and compression are the same problem

★ Turing (1936), *On Computable Numbers*: one universal machine can run every algorithm, and some questions have no algorithm

Turing (1950), *Computing Machinery and Intelligence*: intelligence as behaviour; learning machines proposed

Solomonoff (1964); Kolmogorov (1965): the simplest program that explains data is the best predictor; Occam made precise

Landauer (1961), *Irreversibility and Heat Generation*: erasing a bit has a physical energy cost; computation is physics

Feynman, *Lectures on Computation*: computing from the physics up: reversibility, limits, quantum

MacKay, *Information Theory, Inference, and Learning Algorithms* free: coding, inference and neural nets as one subject

2. Learning theory and statistics

Valiant (1984), *A Theory of the Learnable*: learning becomes a precise computational question (PAC)

Vapnik (1995), *The Nature of Statistical Learning Theory*: capacity, generalisation bounds, structural risk

Wolpert (1996), No Free Lunch: no learner wins on all problems; assumptions are unavoidable

★ Breiman (2001), *Statistical Modeling: The Two Cultures*: prediction vs explanation; why algorithmic models won

Zhang et al. (2017), *Rethinking Generalization*: networks memorise random labels yet generalise on real ones; classical theory is not enough

Belkin et al. (2019), *Reconciling … Bias-Variance*: double descent: past interpolation, bigger can be better

Hastie, Tibshirani, Friedman, *Elements of Statistical Learning* free: the classical toolkit, rigorously

Jaynes, *Probability Theory: The Logic of Science*: probability as extended logic; maximum entropy

3. Neural networks and representation

McCulloch & Pitts (1943): neurons as logic gates: the first formal neural net

Hebb (1949), *The Organization of Behavior*: cells that fire together wire together: learning as local plasticity

Rosenblatt (1958); Minsky & Papert (1969): the perceptron, and the proof of what one layer cannot do

★ Rumelhart, Hinton, Williams (1986): backpropagation learns internal representations

Cybenko (1989): one hidden layer can approximate any continuous function; existence is not learnability

LeCun et al. (1998); Krizhevsky et al. (2012): built-in structure (convolution) plus data plus GPUs

Bengio, Courville, Vincent (2013), *Representation Learning*: what makes a representation good

★ Sutton (2019), *The Bitter Lesson*: methods that scale with compute beat hand-built knowledge

Kaplan et al. (2020), *Scaling Laws*: loss falls as a smooth power law in compute, data, parameters

Olah et al. (2020), *Zoom In* (Distill): networks can be reverse-engineered into circuits

4. Optimisation and decision-making

Robbins & Monro (1951): noisy steps converge: the root of SGD

Bellman (1957), *Dynamic Programming*: optimal decisions decompose recursively

Boyd & Vandenberghe, *Convex Optimization* free: when optimisation is easy, and why duality matters

Sutton & Barto, *Reinforcement Learning* free: learning from reward; temporal-difference learning

Mnih et al. (2015); Silver et al. (2016, 2017): deep RL, then superhuman Go from self-play alone

5. Computers, systems and scale

★ Abelson & Sussman, *SICP* free: programs as abstraction; interpreters all the way down

Moore (1965); Dennard (1974): transistors double; power density stays flat (until it didn't, around 2005)

Amdahl (1967): the serial fraction caps parallel speedup

Lamport (1978), *Time, Clocks, and the Ordering of Events*: in a distributed system, "before" is a partial order

Saltzer, Reed, Clark (1984), *End-to-End Arguments*: put function at the ends; the network stays simple

Dean & Ghemawat (2004), *MapReduce*: commodity clusters as one computer

★ Dean & Barroso (2013), *The Tail at Scale*: at scale, the slowest 1% sets the latency

Jouppi et al. (2017), *TPU*: domain-specific hardware for the dominant workload

Hennessy & Patterson (2019), *A New Golden Age for Computer Architecture*: with Moore and Dennard ending, specialisation is the path

Hennessy & Patterson, *Computer Architecture: A Quantitative Approach*: measure, then design

6. Physics, complexity and emergence

★ Anderson (1972), *More Is Different*: new laws emerge at each scale; reductionism is not construction

Wigner (1960), *Unreasonable Effectiveness of Mathematics*: why math describes nature at all

Jaynes (1957), *Information Theory and Statistical Mechanics*: thermodynamics as inference: max entropy

Hopfield (1982): memory as energy minimisation; physics enters neural nets

Strogatz, *Nonlinear Dynamics and Chaos*: attractors, bifurcations: the language of training dynamics

Feynman, *Lectures on Physics* free: how to think from first principles, at its best

7. Brains and minds

Hubel & Wiesel (1962): the visual cortex is hierarchical feature detectors: the blueprint for CNNs

★ Marr (1982), *Vision*: understand a system at three levels: computation, algorithm, implementation

Dayan & Abbott, *Theoretical Neuroscience*: the brain in equations

8. How breakthroughs happen

★ Hamming (1986), *You and Your Research*: work on important problems; open doors; courage

Hamming, *The Art of Doing Science and Engineering*: learning to learn; style of thinking

Kuhn (1962), *The Structure of Scientific Revolutions*: paradigms, anomalies, and shifts

Pólya (1945), *How to Solve It*: heuristics: specialise, generalise, work backwards

Feynman (1974), *Cargo Cult Science*: the first principle is not to fool yourself

Mathematics, for intuition

3Blue1Brown, *Essence of Linear Algebra / Calculus* free: geometric intuition before symbols

Deisenroth et al., *Mathematics for Machine Learning* free: the working toolkit

How to use this page

One theme per month alongside the LLM roadmap. For each work, ask three questions: *what did it make possible, what did it assume, and where does that assumption break today?* The breakthroughs live in the third answer.